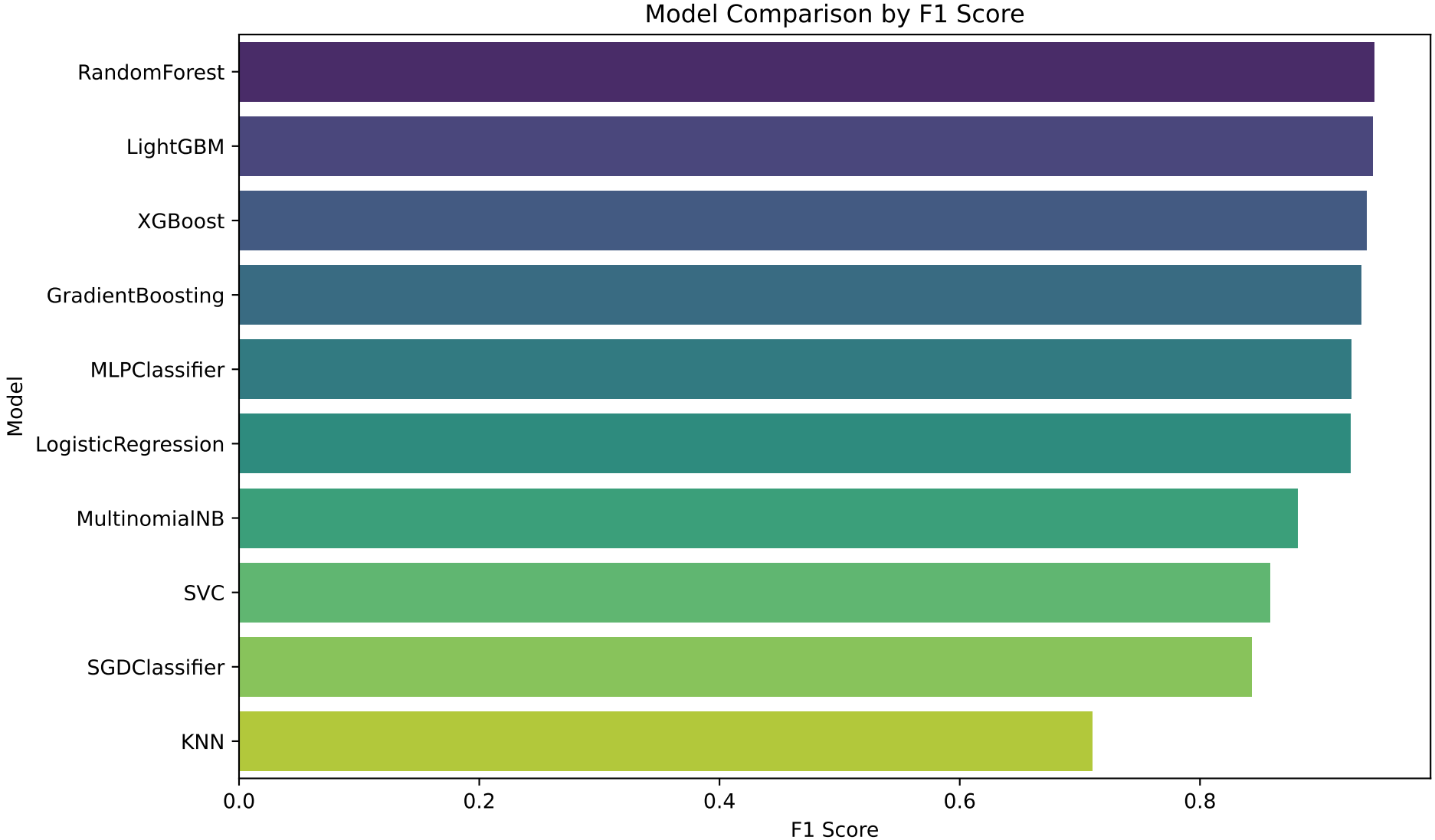
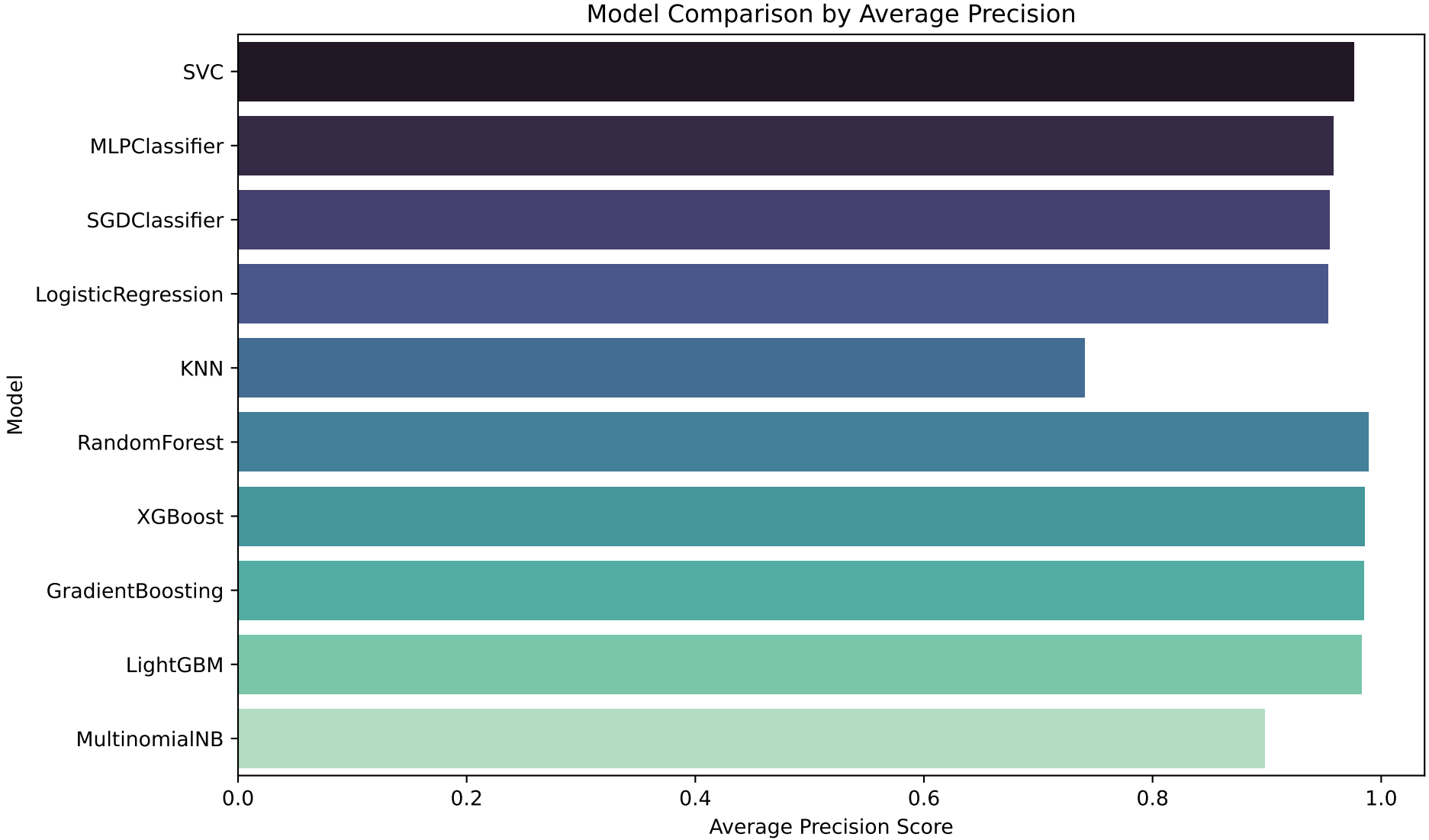
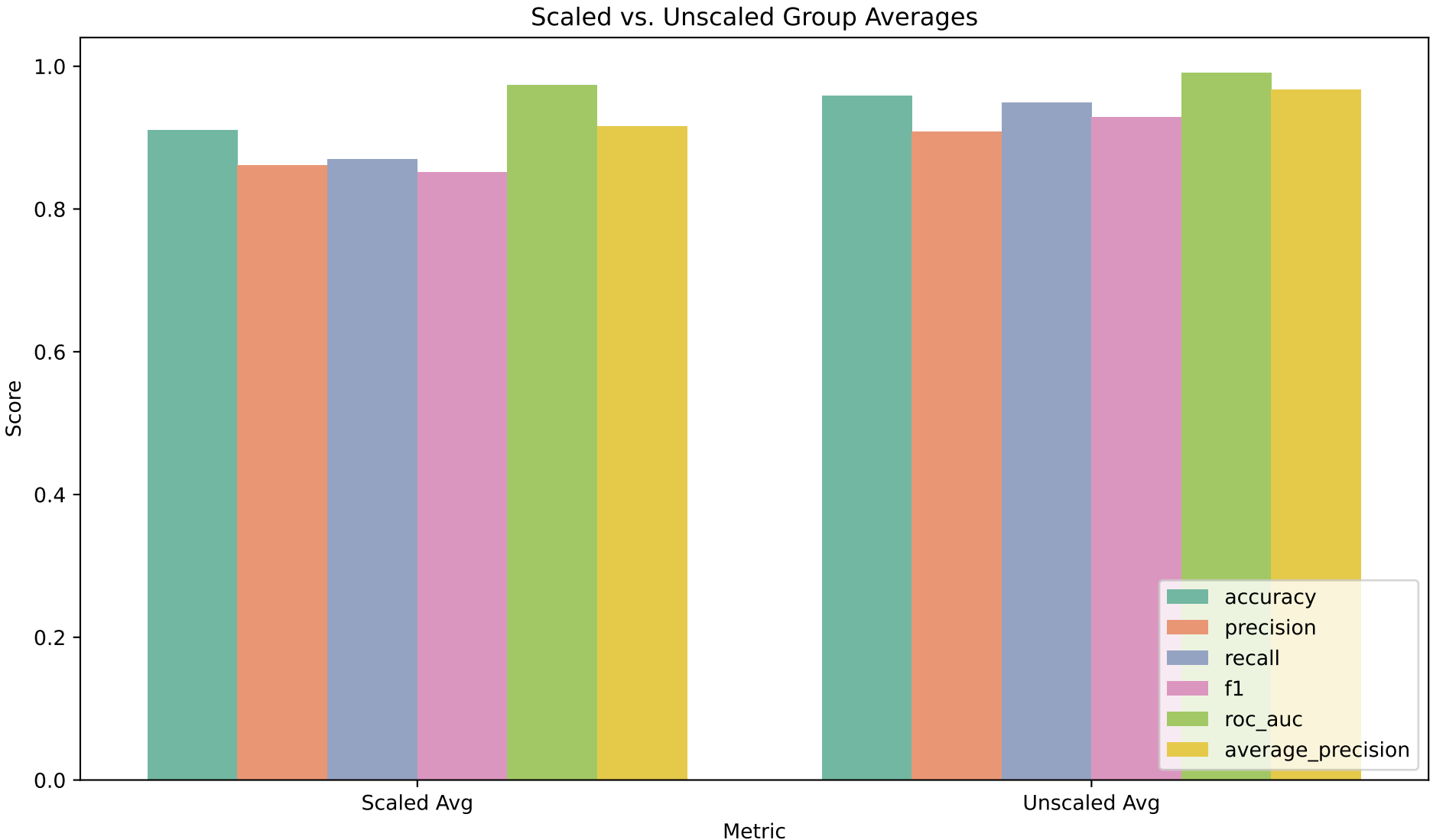
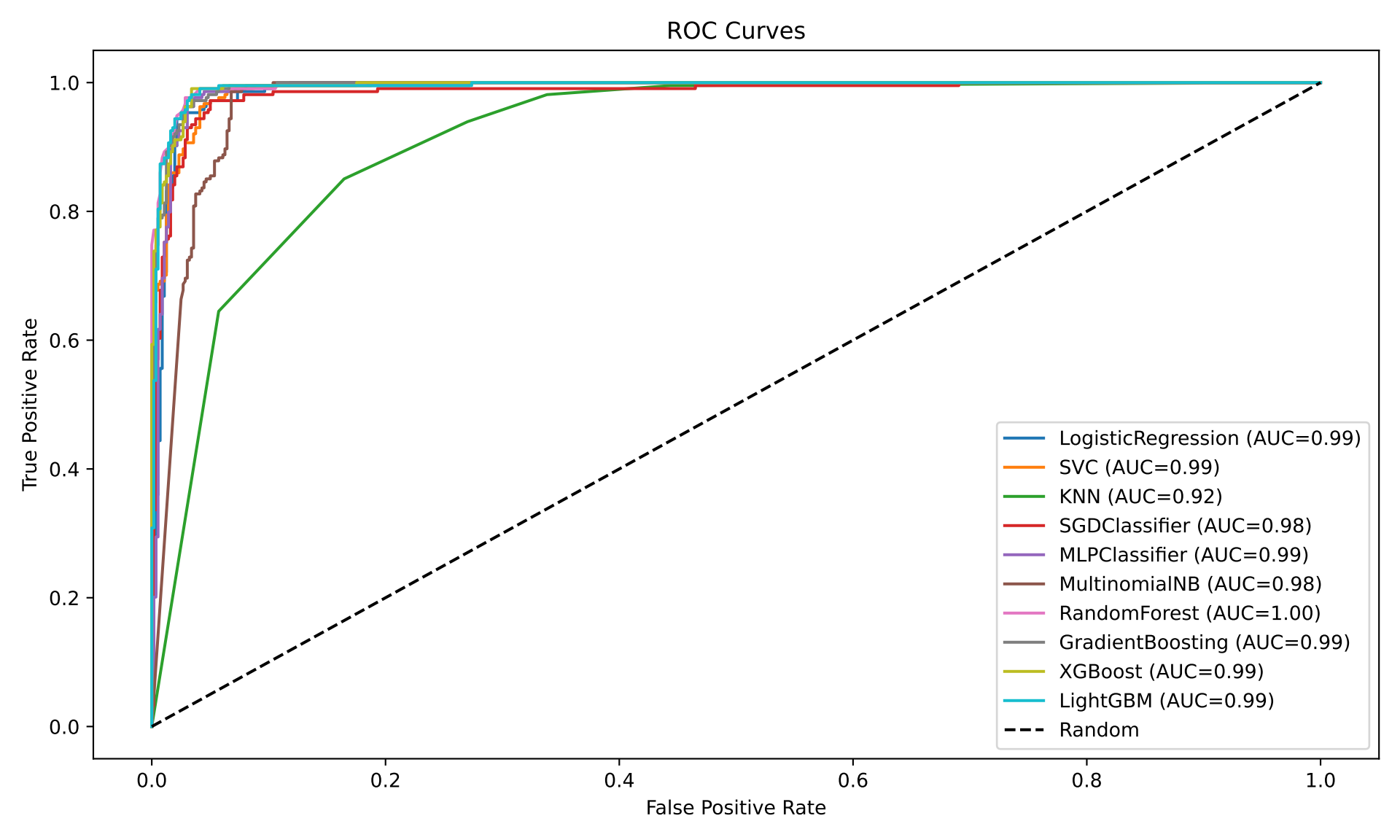
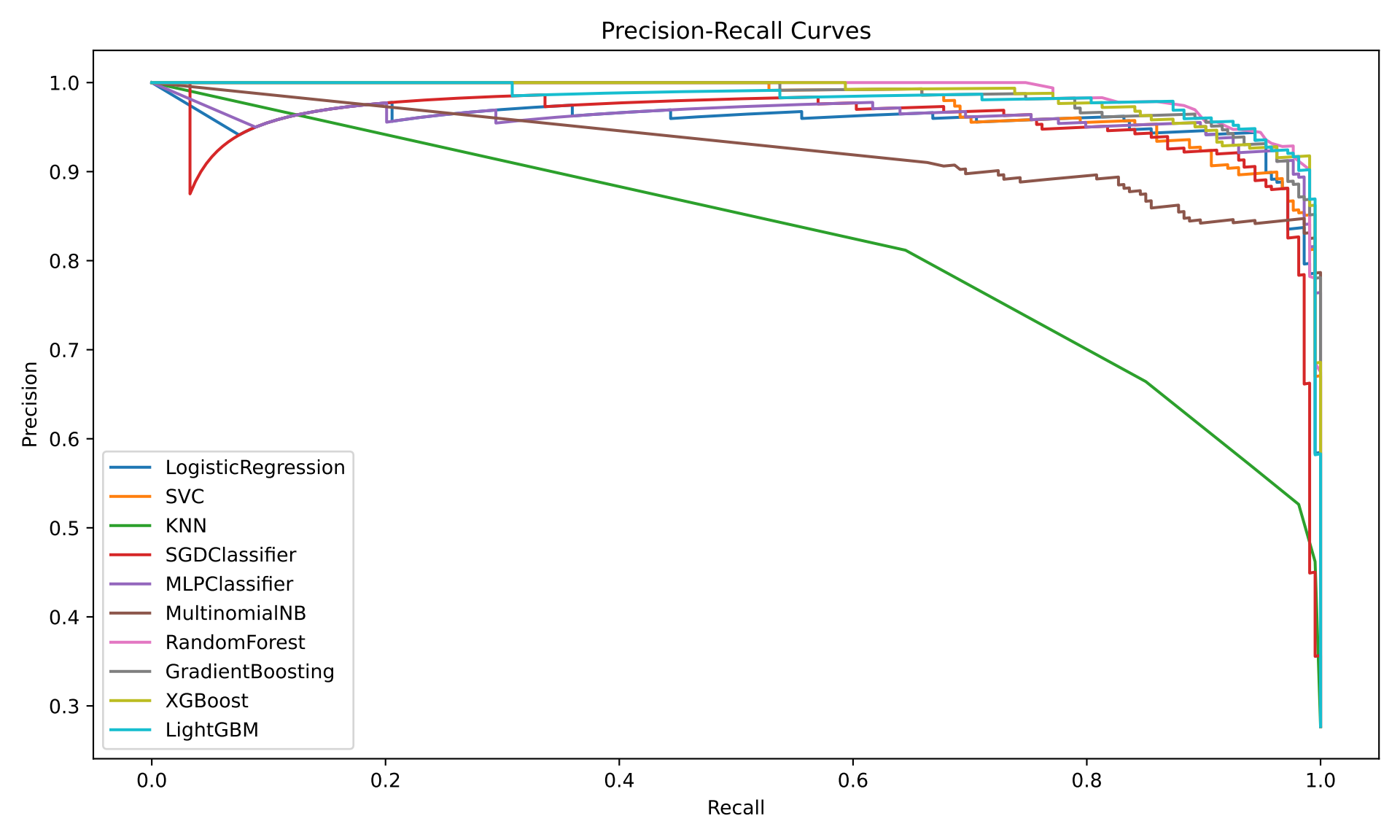

Model Comparison by F1 Score
RandomForest
LightGBM
XGBoost
GradientBoosting
MLPClassifier
LogisticRegression
MultinomialNB
SVC
SGDClassifier
KNN
Model
0.0
0.2
0.4
0.6
0.8
F1 Score

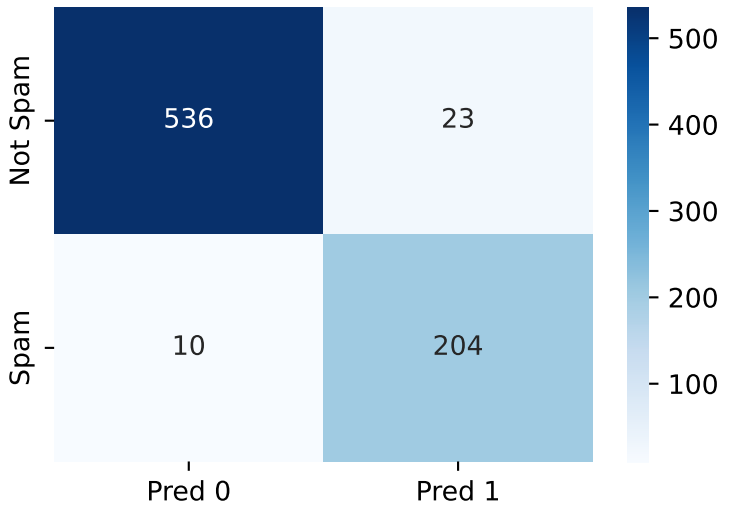

Model Comparison by Average Precision
SVC
MLPClassifier
SGDClassifier
LogisticRegression
KNN
RandomForest
XGBoost
GradientBoosting
LightGBM
MultinomialNB
Model
0.0
0.2
0.4
0.6
0.8
1.0
Average Precision Score


Scaled vs. Unscaled Group Averages
1.0
0.8
0.6
0.4
0.2
0.0
Score
accuracy
precision
recall
f1
roc_auc
average_precision
Scaled Avg
Unscaled Avg
Metric

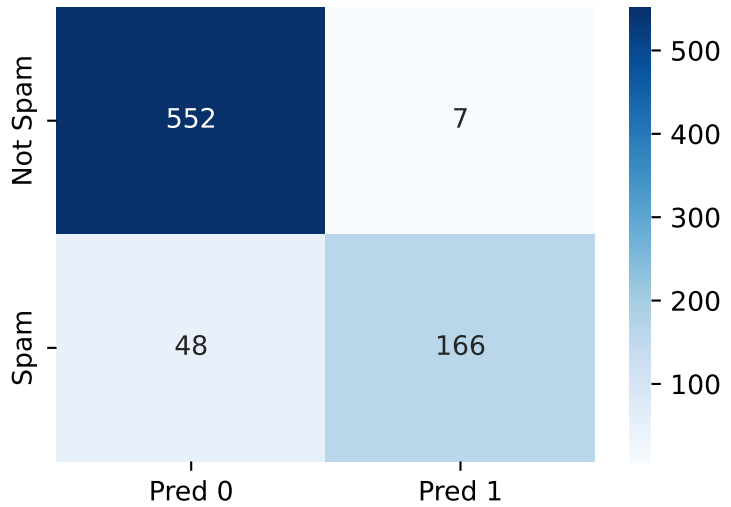

ROC Curves
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
LogisticRegression (AUC=0.99)
SVC (AUC=0.99)
KNN (AUC=0.92)
SGDClassifier (AUC=0.98)
MLPClassifier (AUC=0.99)
MultinomialNB (AUC=0.98)
RandomForest (AUC=1.00)
GradientBoosting (AUC=0.99)
XGBoost (AUC=0.99)
LightGBM (AUC=0.99)
Random


Precision-Recall Curves
Precision
Recall
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.0
0.2
0.4
0.6
0.8
1.0
LogisticRegression
SVC
KNN
SGDClassifier
MLPClassifier
MultinomialNB
RandomForest
GradientBoosting
XGBoost
LightGBM

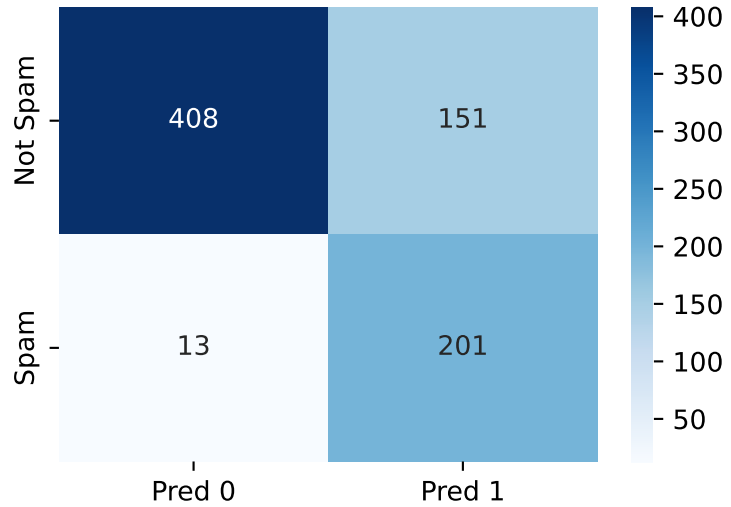
LogisticRegression



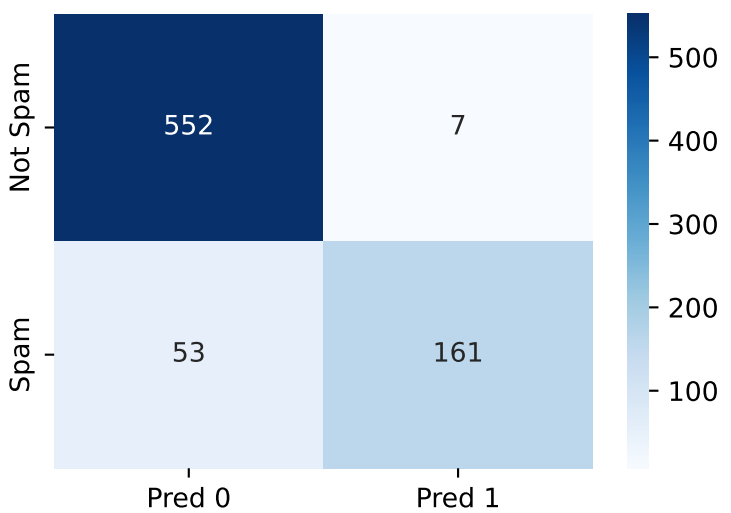
SVC



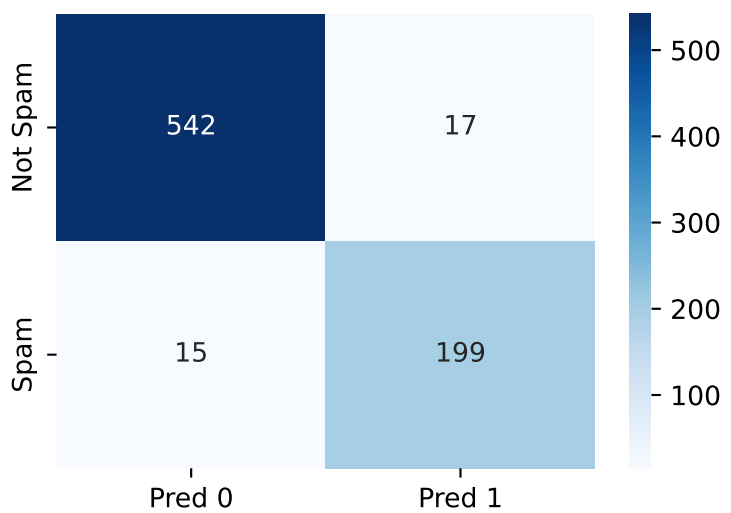
KNN



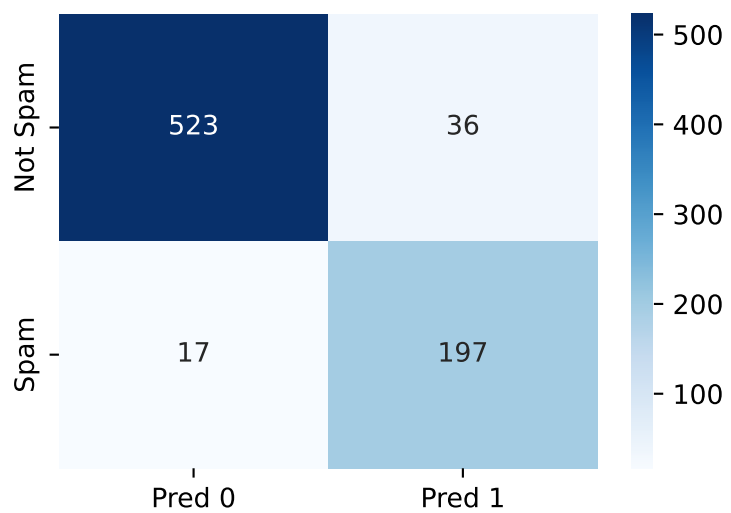
SGDClassifier



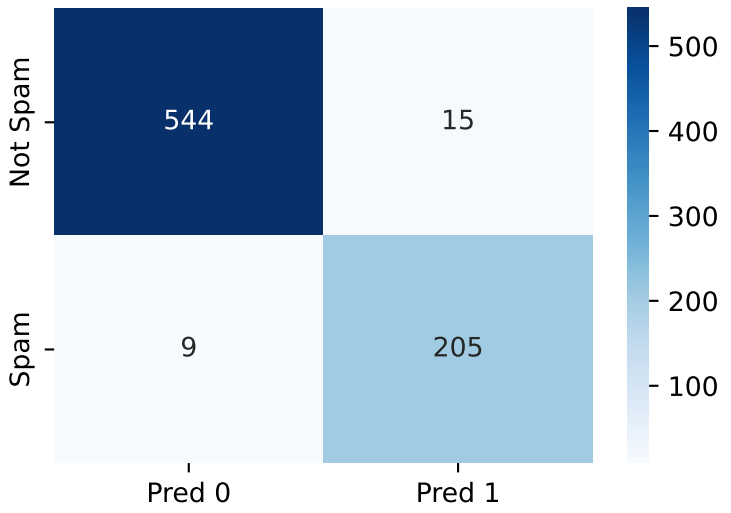
MLPClassifier



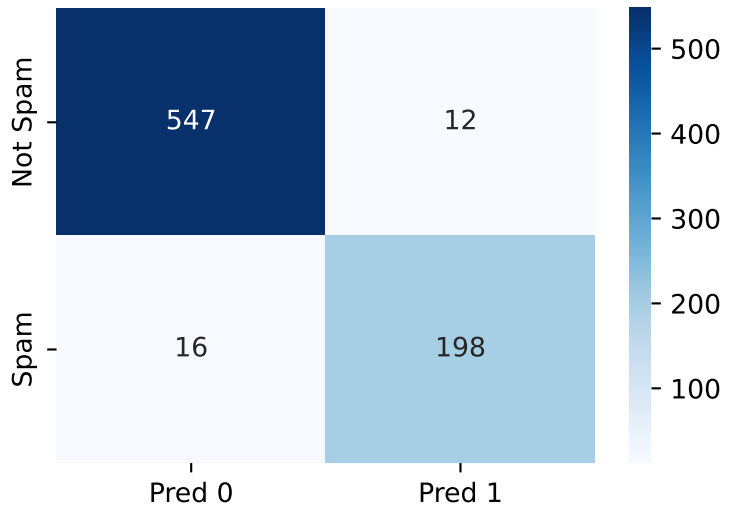
MultinomialNB



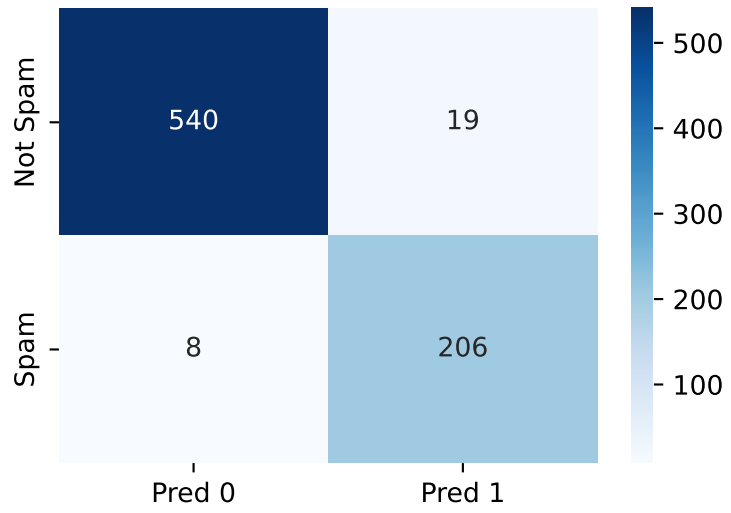
RandomForest



GradientBoosting



XGBoost



LightGBM

